

HSIP Safety Evaluation, Intersection: SR 1602 (Main Street) at SR 1207 (Spring Street)

Order ID	41000076579
Project ID	05-20-62123 (TIP #HS-2005A)
Signal ID	N/A
Location	SR 1602 (Main Street) at SR 1207 (Spring Street)
GPS Coordinates	36.310027, -78.587328
County / City / Division	Granville / Oxford / 5
Countermeasure(s)	Construct a Mini-Roundabout including pedestrian upgrades
Estimated Project Cost	\$278,000
Completion Date	12/15/2021

Treatment Information	Before	After	Percent Reduction (-) / Increase (+)
Total Crashes	7	3	-57%
Total Severity Index	2.06	1.00	-51%
Target Crashes	3	2	-33%
Target Crash Severity Index	1.00	1.00	+0%
Volume (2019, 2025)	5,100	4,300	-16%

Time Period	Start	End	Length
Before Period	3/1/2017	9/30/2021	4y, 7m
Construction Period	10/1/2021	12/31/2021	0y, 3m
After Period	1/1/2022	7/31/2026	4y, 7m

Analysis Criteria
Treatment data consists of all crashes within 150 feet of the intersection (the Y-line) for the before and after periods; construction-period crashes are excluded. Severity index uses the NCDOT EPDO weights (K/A 76.8, B/C 8.4, PDO 1.0).

Target Crashes
Frontal Impact Crashes in Intersection (Angle, LTDR, LTSR, RTDR, RTSR, Head-On)

Crashes Per Year by Project Time Period	Project Development	Before	After
Years	10.00	4.58	4.58
Total	1.90	1.53	0.65
Fatal Injury	0.00	0.00	0.00
Class A Injury	0.00	0.00	0.00
Class B Injury	0.10	0.00	0.00
Class C Injury	0.30	0.22	0.00
Property Damage Only	1.50	1.31	0.44

Injury Crash Summary	Before	After	Percent Reduction (-) / Increase (+)
Fatal Injury Crashes	0	0	n/a
Class A Injury Crashes	0	0	n/a
Class B Injury Crashes	0	0	n/a
Class C Injury Crashes	1	0	-100%
Property Damage Only	6	2	-67%

Target Injury Crash Summary	Before	After	Percent Reduction (-) / Increase (+)
Fatal Injury Crashes	0	0	n/a
Class A Injury Crashes	0	0	n/a
Class B Injury Crashes	0	0	n/a
Class C Injury Crashes	0	0	n/a
Property Damage Only	3	2	-33%

Additional Information	Before	After	Percent Reduction (-) / Increase (+)
Target Crashes: Spring St At-Fault	3	2	-33%
Target Crashes: Main St At-Fault	0	0	n/a
Target Crashes: Failed to Yield	2	2	+0%
Target Crashes: Ran Stop Sign	1	0	-100%
Rear End Crashes	1	0	-100%
On-Street Parking Maneuver Crashes	3	0	-100%

Items for Discussion
- Construction period set as a typical 3 months ending at the 12/15/2021 completion date; Google Earth aerials show the project completed between March 2021 and March 2022 and NearMap shows it in place by February 2022. No crashes occurred at the intersection during the construction period. - All Target Crashes in both periods were failures to stop or yield by drivers on the Spring St approaches; no Main St driver was at fault in either period. - The only injury crash in either period was a before-period Class C rear end at 8:28 PM on 12/6/2019, in which a WB vehicle on E Spring St struck a vehicle stopped at the stop line at Main St. - Before-period Target Crashes: a 12/20/2017 angle crash at 2:10 PM in which an EB vehicle on Spring St ran the stop sign and struck a NB vehicle on Main St; a 7/3/2018 right-turn crash at 5:53 PM in which a SB vehicle turning from Main onto W Spring was struck by a WB through vehicle; and a 5/26/2021 left-turn crash at 11:18 AM in which an EB vehicle turning from Spring onto Main was struck by a WB through vehicle. All three were property damage only. - After-period Target Crashes, both inside the circle and property damage only: a 2/14/2022 angle crash at 9:14 AM in which a WB vehicle on Spring St entered the circle and struck a SB vehicle already in it, and a 1/8/2025 right-turn crash at 9:30 AM in which a vehicle entering from E Spring St through the yield struck a NB vehicle in the circle. A 9/20/2024 sideswipe at 5:23 PM on entry to the circle was the third circle crash, also a failure to yield. - Seven Initial Study crashes were deleted after report review: six backing or parked-vehicle crashes in the Post Office angled parking row on Main St, which begins about 140 ft north of the junction, and one parked vehicle struck mid-block in front of Oxford Baptist Church. Each was coded at the intersection, but the diagram or narrative places it beyond 150 ft. Three before-period parking maneuver crashes in the W Spring St angled spaces at the corner were kept; they are unrelated to the intersection control. 28 additional reports were reviewed and none added: crashes wholly within the Post Office, Library or SunTrust parking lots were excluded, and the rest were at Little John St, Williamsboro St, Orange St, MLK Jr Ave, Park Dr or on rural SR 1602.



East leg: SR 1207 (Spring Street), 35 mph, 2025 AADT 2,800 | West leg: SR 1207 (Spring Street), 35 mph, 2025 AADT 1,900 | North leg: SR 1602 (Main Street), 25 mph, 2025 AADT 2,500 | South leg: SR 1602 (Main Street), 25 mph, 2025 AADT 1,300